

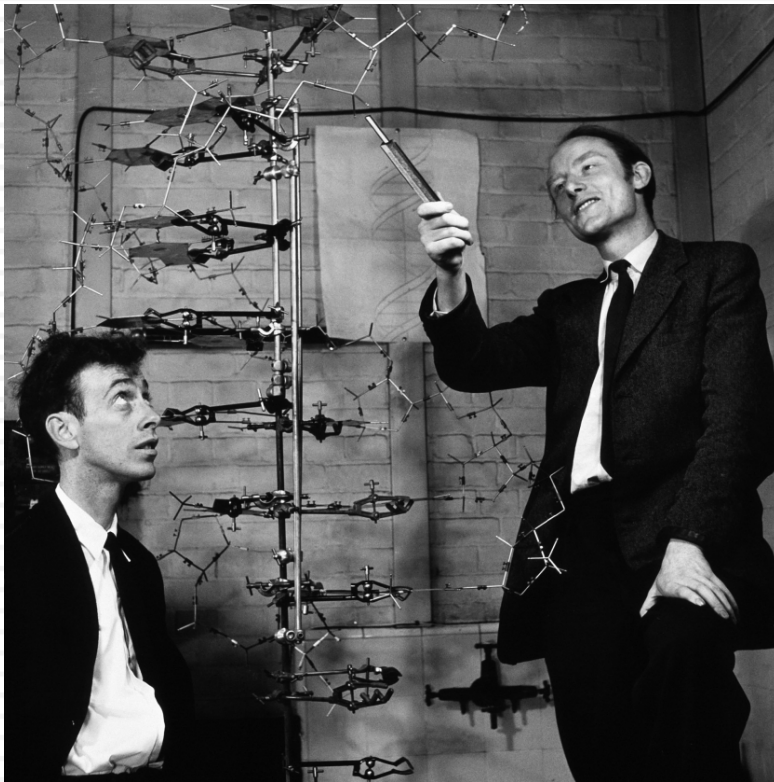
UNIT 4: MOLECULAR GENETICS

4.1 – DNA ADVANCES THROUGH THE 20TH CENTURY

Unit Essential Question

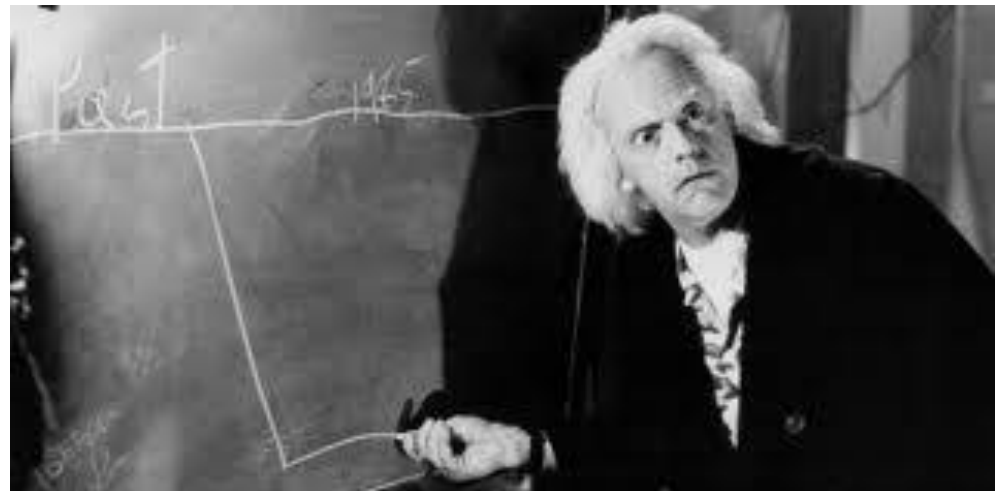
How is genetic information passed through generations?

Part 1 – Historical Basis



Back to the Future

- We will circle back later this unit to discuss Gregor Mendel and his studies that investigated how traits were inherited
 - ▣ Start this unit with understanding studies that followed Mendel into what hereditary material was.
 - ▣ Start small, and zoom out
 - ▣ Understand emergent properties of genetic material from the nucleotide out to heredity.



Historical Basis of Knowledge

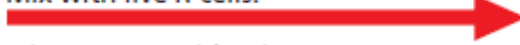
- Following Mendel's studies on heredity in the 1800's scientists through the early 1900's knew certain facts about genetic information.
- Mid-20th century, scientists knew genetic material:
 - Able to *store information* for structure, development, metabolism
 - Stable so it can be *replicated* with high accuracy
 - Able to undergo rare changes called *mutations* to provide genetic variability necessary for evolution
 - *Requirements for genetic material*

Historical Basis - Griffith

Late 1920s England: **Frederick Griffith** studying *S. pneumonia*

- ▣ Causes lung disease pneumonia in mammals
- ▣ Non-infectious strain
- ▣ Infectious (virulent) strain
- ▣ **Virulent strain** surrounded by polysaccharide capsule; protects from body's defenses; grows in colonies with smooth edges (**S strain**)
- ▣ **Non-virulent strain** lacks capsule; grows in colonies with rough edges (**R strain**)
- ▣ Experimented with bacteria in mice to investigate transfer of hereditary information in bacteria

Historical Basis - Griffith

		Result	Conclusion
Experiment 1	 Inject mouse with live <i>R</i> cells. 		<i>R</i> cells do not kill the mouse.
Experiment 2	 Inject mouse with live <i>S</i> cells. 		<i>S</i> cells kill the mouse.
Experiment 3	 Kill <i>S</i> cells with heat.  Inject mouse with heat-killed <i>S</i> cells.		Heat-killed <i>S</i> cells do not kill the mouse.
Experiment 4	 Kill <i>S</i> cells with heat. Mix with live <i>R</i> cells.  Inject mouse with mixture.		Hereditary material from the heat-killed <i>S</i> cells transforms <i>R</i> cells. The transformed <i>R</i> cells kill the mouse.

Griffith's conclusions:

Heat-killed virulent bacteria released a **hereditary factor** that transfers disease-causing ability to live, previously non-virulent bacteria

Transfer of genetic material from one cell to another; one organism to another:
transformation

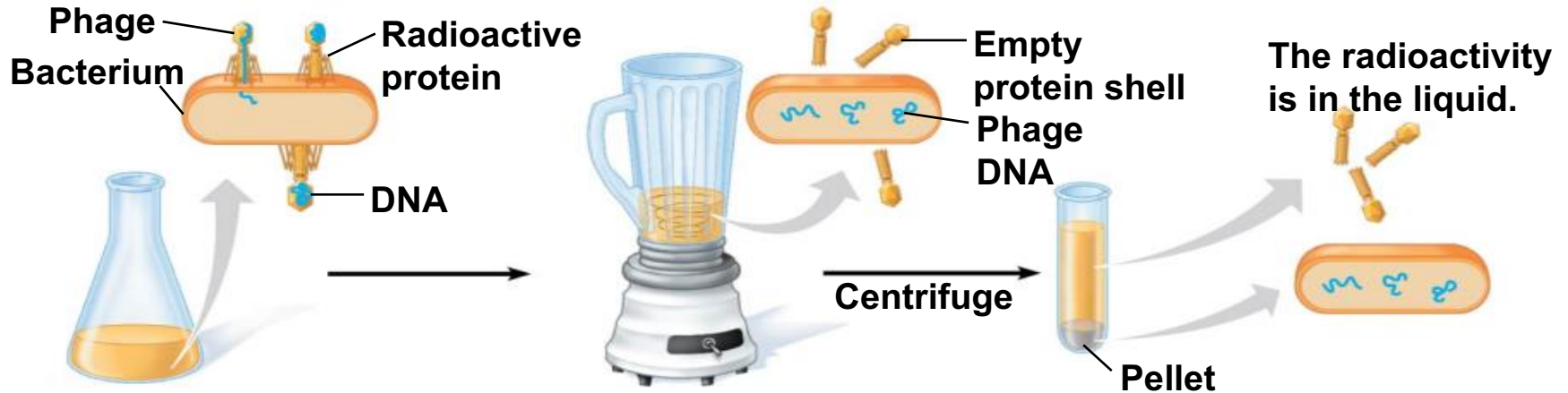
Historical Basis - Avery

- **Early 1940's America:** Oswald Avery sought to determine whether **transforming agent** from Griffith's experiment was **protein, RNA or DNA**
- **Enzymes** were used to destroy each of the 3 molecules in heat-killed S cells
 - ▣ **Protease:** destroys **protein** (experiment 1)
 - ▣ **RNase:** destroys **RNA** (experiment 2)
 - ▣ **DNase:** destroys **DNA** (experiment 3)
- Mixed experimental batches of heat-killed S cells with live R cells and injected into mice
- **Findings:**
 - ▣ Cells missing protein and RNA were still able to transform R cells: mice died
 - ▣ Cells missing DNA could NOT transform R cells: mice lived
- **Conclusion:** **DNA** is the molecule **responsible** for **transformation of bacteria**

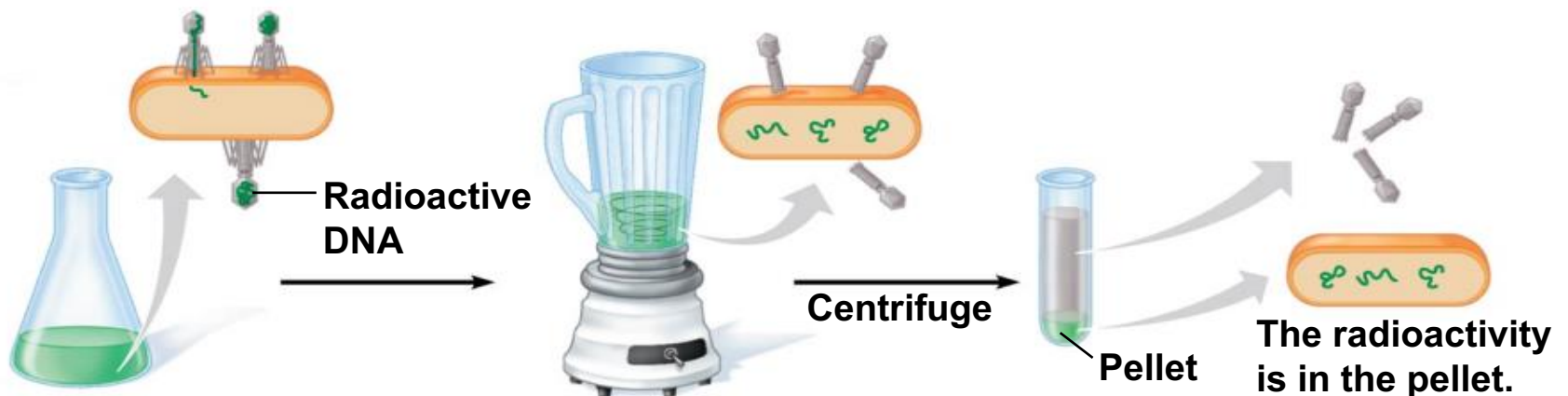
Historical Basis – Hershey-Chase

- **Early 1950's America: Chase & Hershey** sought to **determine** whether **DNA or protein** was **hereditary material** transferred by viruses entering a bacterium
 - ▣ **Bacteriophage (phage)**: virus that infects bacteria
- **Grew phage** in presence of **radioactive sulfur** (^{35}S) to label **proteins**
- **Grew phage** in presence of **radioactive phosphorus** (^{32}P) to label **DNA**
- Allowed labeled phage to infect bacteria
- Used a blender to remove phage coats from cells
- Used a centrifuge to separate phage from bacteria
- **Findings:**
 - ▣ **Radioactive sulfur** present in **supernatant (viruses)** not bacterial **cell pellet**
 - ▣ **Radioactive phosphorus** present in **bacterial cell pellet, not supernatant**
- **Conclusion:**
 - ▣ **DNA** is the **hereditary material** of **viruses**
- [Hershey Chase Visualized](#)

Historical Basis – Hershey-Chase



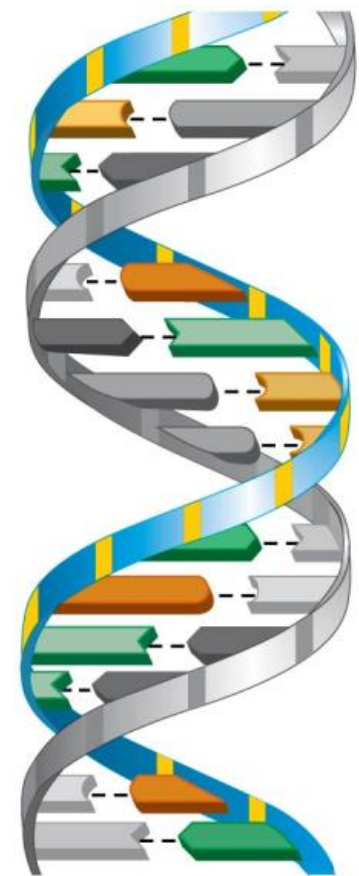
Batch 1: Radioactive protein labeled in yellow



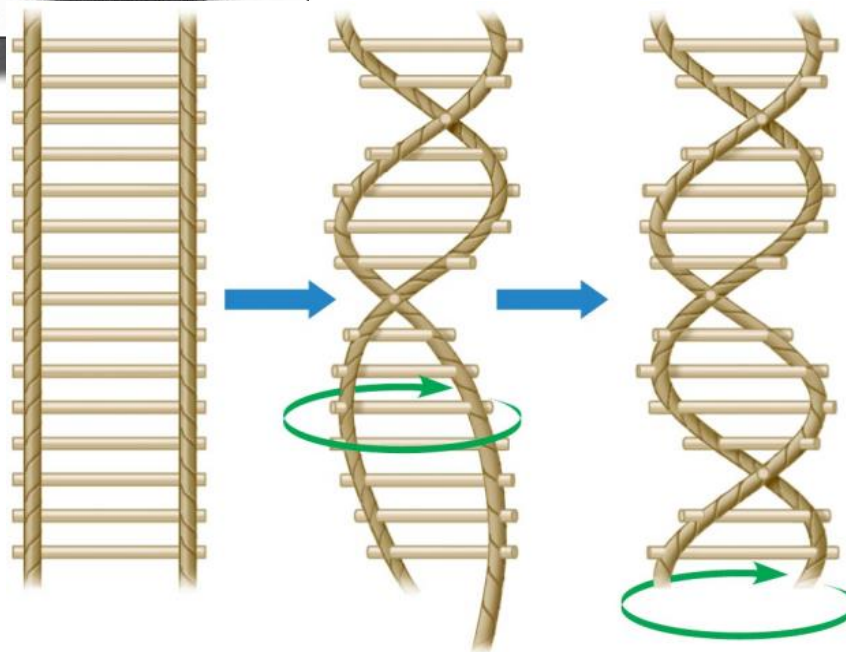
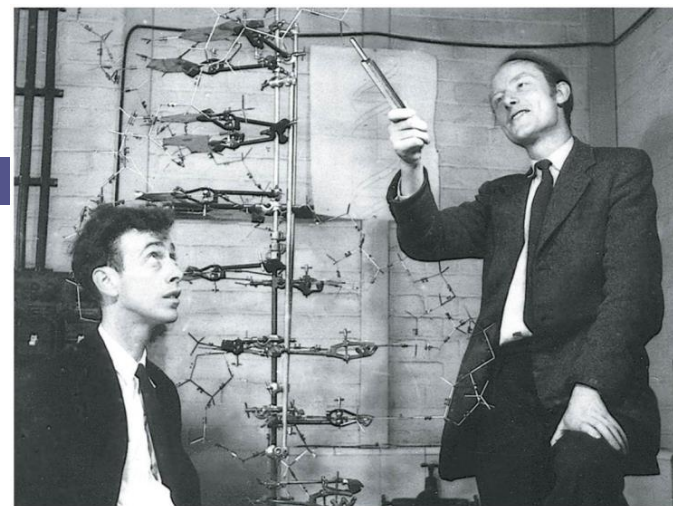
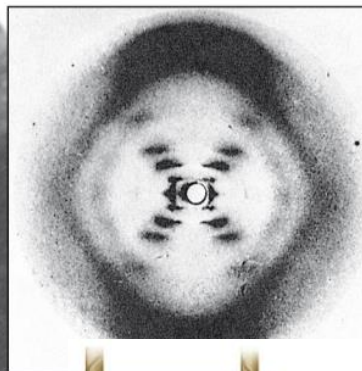
Batch 2: Radioactive DNA labeled in green

Historical Basis – Watson, Crick, Wilkins, & Franklin

- After DNA was determined to be hereditary material – next challenge: **determine its structure**
- Maurice **Wilkins**, Rosalind **Franklin** **purified** and **crystallized DNA**, used a technique called **X-ray diffraction** to image it
- James Watson, Francis Crick used this information to propose a model for DNA structure in 1953
 - ▣ **Publication in Nature: 1 page** long with **1 hand drawn image** of the structure
 - ▣ Won **Nobel Prize** in Medicine, **1963**, along with **Wilkins** for the discovery
 - ▣ **Franklin died** from radiation exposure **before award was given**, but is recognized for producing the famous image that led to the discovery of the structure
- Proposed structure:
 - ▣ **2 chains** wrapped around each other to form a **double helix** (winding, spiral staircase)
 - ▣ Also proposed a **mechanism for replication** of DNA



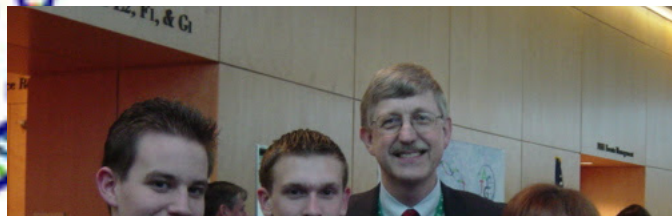
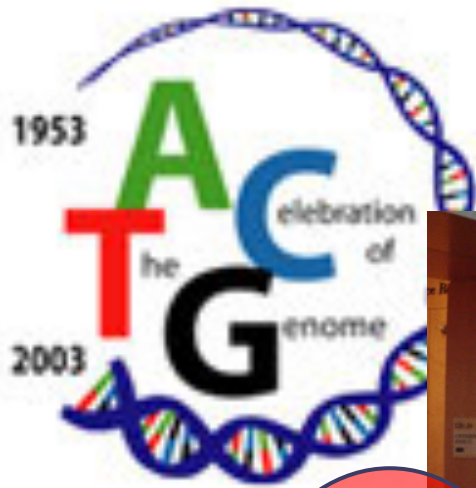
A DNA
double helix



Twist

50 Years Later

April 2003, Conference announcing the completion of the
Human Genome Project
Bethesda, MD, National Institute of Health (NIH)



Lesson Takeaways

1. Take a look at the papers from the studies we looked at today – Available in Classroom
 - ▣ Write in your lab notebook a few observations between these historical papers compared to papers you have read for Journal Club & ITR
 - Structure, look, tone, etc
2. Read the Watson & Crick Paper – In Classroom
 - ▣ Take note as you read simplicity in tone, and information presented. Won them the Nobel Prize!
- ▣ Go on the Jamboard in Classroom and fill in a sticky note today